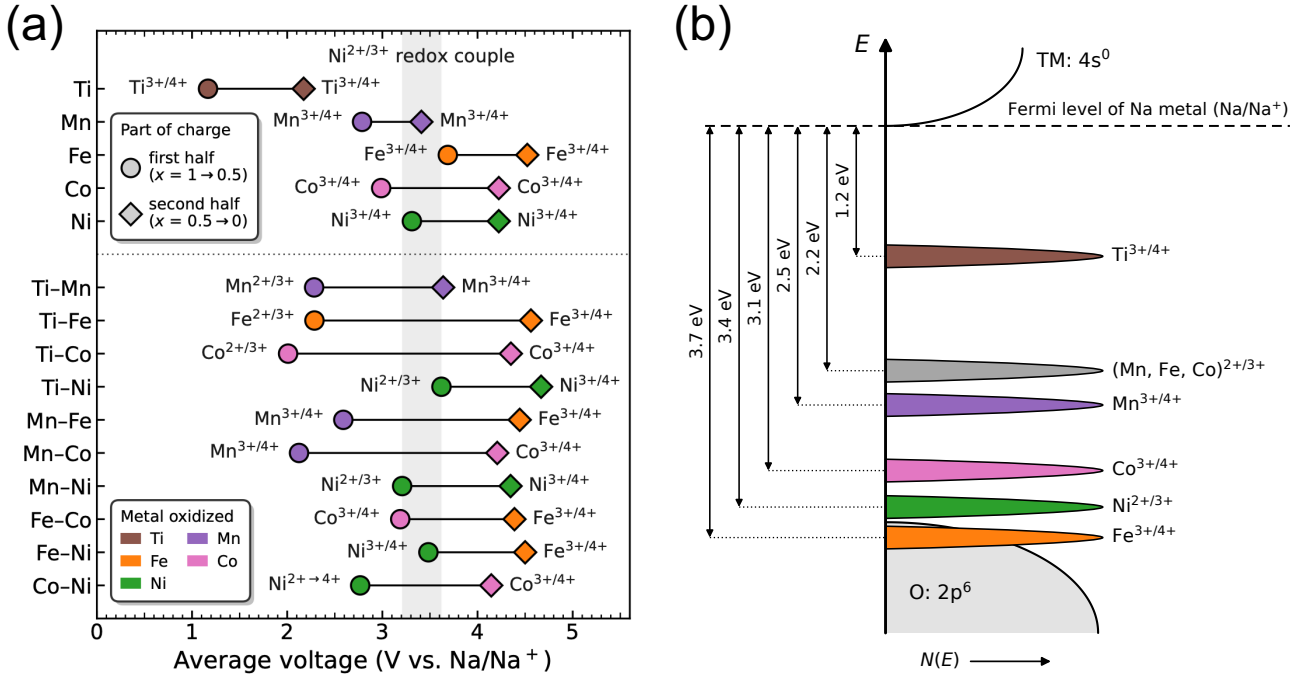

(a)
Ni2+/3+ redox couple
Ti
Mn
Fe
Co
Ni
Ti3+/4+
Mn3+/4+
Fe3+/4+
Co3+/4+
Ni3+/4+
Part of charge
first half (x = 1 → 0.5)
second half (x = 0.5 → 0)
Ti–Mn
Ti–Fe
Ti–Co
Ti–Ni
Mn–Fe
Mn–Co
Mn–Ni
Fe–Co
Fe–Ni
Co–Ni
Mn2+/3+
Fe2+/3+
Co2+/3+
Ni2+/3+
Ni2+→4+
Metal oxidized
Ti
Mn
Fe
Co
Ni
0
1
2
3
4
5
Average voltage (V vs. Na/Na+)
(b)
E
TM: 4s0
Fermi level of Na metal (Na/Na+)
1.2 eV
2.2 eV
2.5 eV
3.1 eV
3.4 eV
3.7 eV
Ti3+/4+
(Mn, Fe, Co)2+/3+
Mn3+/4+
Co3+/4+
Ni2+/3+
Fe3+/4+
O: 2p6
N(E)